ECONOMY 15 | FOOD PROCESSING, COLD CHAINS & VALUE ADDITION

01. CHAIN THESIS

Farm -> aggregate -> condition -> process -> standards -> market.

Plant construction alone does not create a value chain.

FACT -> MECHANISM -> QUALIFICATION

02. PROCESSING LEVELS

Primary conditioning -> secondary manufacture -> tertiary convenience.

Value added differs from sale value.

FACT -> MECHANISM -> QUALIFICATION

03. VALUE CAPTURE

Longer shelf life + quality + demand may raise value.

Contracts and competition decide the farmer share.

FACT -> MECHANISM -> QUALIFICATION

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04. LOSS / WASTE / NUTRITION

Food loss is mainly upstream; waste mainly retail/consumer.

Processing has product-specific nutritional effects.

FACT -> MECHANISM -> QUALIFICATION

05. COLD CONTINUITY

Pre-cool -> packhouse -> storage -> reefer -> last mile.

Cold storage alone is not a chain.

FACT -> MECHANISM -> QUALIFICATION

06. TEMPERATURE CHAINS

Chilled != frozen != CA/MA.

Dairy, fish, fruit and frozen foods need different protocols.

FACT -> MECHANISM -> QUALIFICATION

07. BOTTLENECKS

08. REGULATORS

09. PMKSY / PMFME

FACT -> MECHANISM -> QUALIFICATION

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07. BOTTLENECKS

Seasonality + small lots + working capital + power + skills.

Sanctioned capacity != operating throughput.

FACT -> MECHANISM -> QUALIFICATION

08. REGULATORS

FSSAI: FSS Act 2006 domestic safety.

APEDA: APEDA Act 1985 scheduled-product exports.

FACT -> MECHANISM -> QUALIFICATION

09. PMKSY / PMFME

PMKSY Rs 6,520 crore through 2025-26.

PMFME Rs 10,000 crore through 2025-26.

FACT -> MECHANISM -> QUALIFICATION

10. MFP / OPERATION GREENS

11. PLI / AIF / FDI

12. REFORM VERDICT

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10. MFP / OPERATION GREENS

Mega Food Parks: no new approvals from 1 April 2021.

Operation Greens: TOP origin; notified expansion.

FACT -> MECHANISM -> QUALIFICATION

11. PLI / AIF / FDI

PLISFPI Rs 10,900 crore through 2026-27.

AIF debt support; processing automatic-route FDI.

FACT -> MECHANISM -> QUALIFICATION

12. REFORM VERDICT

Infrastructure + institutions + standards + credit + skills + market.

Audit throughput, farmer returns, jobs and safety.

FACT -> MECHANISM -> QUALIFICATION